

Vinh Ly

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Summary

Software Engineer with **18 years** of experience building systems that serve billions of users. Built search ranking infrastructure at Google, scaled **YouTube** video delivery to millions of concurrent viewers, and designed **Whatnot** marketplace handling **10M+** daily transactions with **C++** and **Go**. Owned end-to-end reliability for **GCP** microservices on **Kubernetes**, led teams of 10 engineers through complex organizational challenges, and managed on-call rotations across three companies. Next focus: autonomous vehicle platform systems with geospatial data pipelines and large-scale distributed compute. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Go, Python, Java, SQL, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, GCP, distributed systems, service mesh, Linux

Domains: backend services, data pipelines, search systems, real-time systems, ML infrastructure

Leadership: technical strategy, team management, architecture design, on-call ownership, Postgres, AWS

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Design video infrastructure services handling millions of concurrent streams with **C++** and **Go**, focusing on low-latency delivery and system reliability across **GCP**.
- Own end-to-end deployment pipelines for core playback services, reducing time-to-production for critical fixes by 30% through improved tooling and automation.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Mentored team of 5 engineers on distributed systems design patterns and **C++** best practices, resulting in faster code reviews and reduced bug rates.
- Established on-call rotation and incident response protocols for platform services, improving mean time to resolution by 25% within the first quarter.
- Built real-time auction infrastructure processing 100k+ concurrent connections, reducing p99 latency by 40% through service mesh optimization and caching strategies.
- Delivered marketplace features end-to-end from design review through production deployment, serving **10M+** daily transactions across **C++** and **Go** microservices.
- Reduced cloud infrastructure costs by 15% through Kubernetes pod optimization and resource right-sizing across **GCP**.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers building core search infrastructure serving billions of queries daily, delivering improvements to ranking relevance and system availability.
- Designed distributed caching layer that reduced backend load by 35% while maintaining sub-10ms response times for top search queries.
- Led technical strategy for indexing pipeline processing 100TB+ daily, improving freshness of search results by 20% through pipeline parallelization.
- Established engineering career development framework including promotion calibration processes and mentorship programs for junior and senior engineers.
- Coordinated cross-team initiatives spanning search, ads, and infrastructure organizations to ship features affecting all Google users.

Software Engineer | Google | 02/2011 - 02/2015

- Built **C++** search ranking components that improved relevance metrics by 12% across major query categories through feature engineering and model tuning.
- Collaborated with research scientists to transition machine learned models from prototype to production, handling 50M+ queries per day.
- Developed **Python** automation tooling eliminating manual review processes, saving 10+ engineering hours weekly across the team.
- Participated in on-call rotation for search serving infrastructure, resolving production incidents affecting query response quality.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Delivered features for developer tools on schedule, gathering customer feedback and prioritizing improvements based on usage data.
- Improved automated test coverage from 70% to 92% through test-driven development adoption and legacy code refactoring.
- Mentored two intern engineers who received full-time offers, one joining the team directly after graduation.
- Reduced build times by 25% through parallelization improvements and optimized compilation pipelines.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented reporting feature for enterprise management console in **C++**, shipped to customers in the fall release.
- Automated regression testing workflow, cutting manual QA cycles from 3 days to 4 hours.

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007